

Learning outcomes	Additional guidance
<i>Through use of the apparatus and techniques listed below, and a minimum of 12 assessed practicals (see Section 5g), learners should be able to demonstrate all of the practical skills listed within 1.2.1 and CPAC (Section 5g, Table 2) as exemplified through:</i>	
(a) use of appropriate apparatus to record a range of quantitative measurements (to include mass, time, volume, temperature, length and pH)	HSW4
(b) use of appropriate instrumentation to record quantitative measurements, such as a colorimeter or potometer	HSW4
(c) use of laboratory glassware apparatus for a variety of experimental techniques to include serial dilutions	HSW4
(d) use of a light microscope at high power and low power, including use of a graticule	HSW4
(e) production of scientific drawings from observations with annotations	HSW8
(f) use of qualitative reagents to identify biological molecules	HSW4
(g) separation of biological compounds using thin layer/paper chromatography or electrophoresis	HSW4
(h) safe and ethical use of organisms to measure: (i) plant or animal responses (ii) physiological functions	HSW4, HSW10
(i) use of microbiological aseptic techniques, including the use of agar plates and broth	HSW4
(j) safe use of instruments for dissection of an animal or plant organ	HSW4
(k) use of sampling techniques in fieldwork	HSW4
(l) use of ICT such as computer modelling, or a data logger to collect data, or use of software to process data.	HSW3, HSW4